

Yoen Zhang

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EDUCATION

University of Waterloo | Bachelor of Computer Science (Honours)

Sep 2020 - Apr 2025 | Waterloo, ON

TECHNICAL SKILLS & TOOLS

Languages: TypeScript, JavaScript, Python, SQL, Swift, Java, HTML/CSS

Frameworks/Libraries: React, Next.js, Node.js, Express.js, Tailwind CSS, SwiftUI, Flask

AI & ML: Claude Code, Cursor, Conductor, MCP, Skill Libraries, Agent Plugins, PyTorch, scikit-learn, FAISS

Tools & Platforms: AWS, GCP, Git, Docker, CI/CD, PostgreSQL, Supabase, MongoDB, Airflow, Figma, Framer

EXPERIENCE

Founding Solutions AI Engineer | Velt

Remote - San Francisco, CA | Aug 2025 - Present

- Architected **3+ agentic AI pipelines (6-12 agent chains each)** replacing entire dev workflows: **10x faster** documentation publishing, **8x faster** client SDK installation, & **5x faster** sample app creation
- Deployed **50+ sample apps** in **TypeScript, React, & Next.js** covering every SDK feature, equipping **GTM & solutions teams** with zero-friction demos to accelerate prospect evaluation, product fit & contract conversion
- Shipped **14+ agent skill libraries, 2 MCP servers, 2 agent plugins, & a CLI** enabling clients to integrate the Velt SDK hands-free into any codebase, including an **agentic pipeline** that auto-updates the skill libraries to Anthropic's latest standards
- Designed & built an end-to-end **Playwright** testing framework covering every major SDK feature, shipping **1,100+ tests in 10 days** via an AI pipeline that writes, manages, & tracks test suites; deployed on **GCP** with **GitHub Actions** triggers running on every SDK production release
- Led **team-wide AI adoption**, coaching peers on agent-chain workflows; engineered an internal agent chain that turns integration requests into production-ready **Velt adapters** for public libraries like **Tiptap & Yjs**, cutting dev time from **2 days to 40 min**

Mobile Software Developer - iOS | ESPN Bet, PENN Entertainment

Toronto, ON | Jan 2024 - Aug 2024, May 2025 - Aug 2025

- Completed **160+ tickets** over **12 months**; recognized as the **top-performing intern** company-wide over a **1-year tenure**
- Led iOS development for the "**Parlay Lounge**", "**BetVision LiveStream**", & "**MyBets 3.0**" using **SwiftUI**, collaborating with product & design to launch **4 weeks ahead of schedule**, improving roadmap velocity
- Developed & shipped flagship UI features, utilizing **Swift & UIKit**, driving engagement across **2.5M+ monthly users**
- Refactored Casino-native & legacy components into **modular, reusable SwiftUI packages**, enabling **cross-team adoption** & improving code consistency across the mobile platform

Data Pipeline Engineer – Undergraduate Research Assistant | TRAFFIC

Remote - London, UK | May 2025 - Aug 2025

- Designed and deployed an end-to-end **Airflow** pipeline to ingest, deduplicate, and classify ivory-trafficking news from six sources (Google Alerts, NewsAPI, NYTimes RSS, Currents API, Guardian API, Mediastack API), achieving comprehensive global coverage
- Upgraded the **NLP extraction module** from **GPT-3.5** to **Gemini 2.5 Flash Pro** for structured article parsing; implemented JSON parsing with fallback mechanisms and intelligent negative-example collection, reducing false positives by **70%**
- Engineered an **ML pipeline** with **TF-IDF vectorization**, a **Random Forest classifier**, and optimal thresholding; achieved **99.9% accuracy, 100% precision, and 99.7% recall** on **6,000+ examples**, a **25% F1 gain** over the baseline Logistic Regression model
- Built robust **data scraping and enrichment** pipelines using **Python, BeautifulSoup, and PostgreSQL**, incorporating retry logic, idempotent task execution, and comprehensive error handling across multiple API integrations

Frontend Developer | Mozilla

Toronto, ON | Sep 2022 - Dec 2022, May 2023 - Aug 2023

- Designed & managed **A/B experiments** enrolling **10,000-1,000,000+ users**, influencing key **Firefox** features, **product strategy** & **user experience** decisions for millions of users worldwide
- Integrated backend telemetry with **Glean SDK**, contributing to large-scale data aggregation in cross-platform experiments
- Developed modular & reusable **React** components adopted by **5+ engineering teams**, standardizing UI patterns company-wide
- Upgraded an internal **SCRUM tracking tool** using **React** & integrated custom tagging to replace manual JIRA processes, improving bug reporting & increasing team ticket output by **200%**
- Developed automated **Node.js** scripts to populate **Looker dashboards**, reducing manual reporting time for internal stakeholders

Fullstack/Backend Developer | Mobials Inc

Toronto, ON | Jan 2022 - Apr 2022

- Enhanced **AutoVerify's API** using **PHP** & architected **RESTful APIs** for partner integrations, allowing **1,000+ business clients** to offer real-time pre-approvals for insurance & vehicle estimates
- Optimized **SQL queries** by **50%**, leveraging **Postman** for API testing to improve accuracy & system efficiency
- Automated PHPUnit tests & CI/CD pipelines** for backend services, reducing deployment bugs by **40%**

Fullstack Developer | Goodfood Market Corp

Toronto, ON | May 2021 - Aug 2021

- Developed **full-stack data anonymization scripts** using **Laravel** & a **Vue.js** interface, allowing internal teams to securely hash or remove customer data based on subscription changes
- Migrated & optimized **over 1.5 million user records** and **10+ million data points** using **MySQL queries**, ensuring alignment with evolving business models & subscription plans

PERSONAL PROJECTS

Suggestify | github.com/yoenzhang/Suggestify | Python, PyTorch, React Native, Flask, FAISS, MFCCs

- Designed an AI-powered music recommendation app inspired by **Shazam**, leveraging **MFCC-based audio fingerprints & FAISS indexing** to match user-input audio against a database of **8,000+ songs**
- Built a **Flask** backend with a **React Native** front-end, enabling real-time song identification & AI-driven suggestions

StatCheck | github.com/yoenzhang/StatCheckWeb | Python, Swift, React, NBA API

- Developed an iOS app using **Swift & React**, integrating the **balldontlie.io API** for real-time NBA statistics dating back to 1946